EEG RESEARCH · WORD-READING DETECTION

Detecting Word-Reading Events from Brain Signals

A plain-language account of how we built, tested, and combined two independent methods for spotting the moment a word is being read, using nothing but raw EEG — and how we verified the results without eye-tracking data at inference time.

Dataset: ZuCo1 (12 subjects, Normal Reading + Task-Specific Reading), plus a single self-contained validation block for a second subject (YHS)

Ground truth source: eye-tracking-derived word onset times, recovered independently from the raw EEG's own event markers

Methods covered: VWFA envelope detection, Wernicke-area ERP detection, and their combination

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What we were trying to do

The goal is simple to state and hard to do: looking only at a stream of raw EEG (electrical brain activity recorded from the scalp), can we tell *exactly when* a person is reading a specific word — without using their eye movements at all?

Normally, researchers know when someone is reading a word because they track where the person's eyes are pointing (eye-tracking). That works well, but it needs extra hardware and doesn't tell you anything about what's happening inside the brain. We wanted to see whether the brain signal itself carries enough information to reconstruct that same timing — which would mean, in principle, word-by-word reading could be tracked from EEG alone.

To do that fairly, we needed two things: (1) one or more detectors that only look at EEG and guess "a word is being read right now," and (2) an independent, trustworthy answer key ("ground truth") saying when words were *actually* read, so we can grade the detectors. Building that answer key turned out to be its own small project, described in Section 2.

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Building the answer key: how do we know when a word was really read?

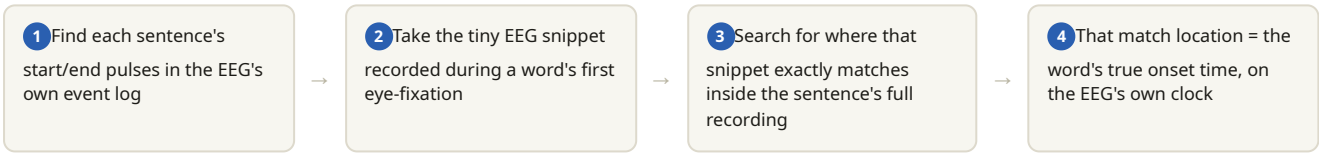
The EEG recordings we used come from a public dataset (ZuCo) where people read real sentences while wearing both an EEG cap and an eye-tracker. The eye-tracker data tells us, precisely, when someone's eyes landed on each word and for how long. That's our ground truth — but there's a catch: the eye-tracker and the EEG recorder are two separate machines with two separate clocks, and by default there's no simple formula to line up "eye-tracker time" with "EEG time."

> *Think of it like having two stopwatches that were started at slightly different moments, with no shared "zero." You can't just compare their readings directly.*

The trick: match the actual signal, not the clock

Instead of trying to synchronize the two clocks, we used a more reliable shortcut. The dataset's raw EEG file already contains, hidden inside its own event log, brief pulses (called "triggers") that mark exactly when each sentence started and ended — on the EEG's own timeline, with no ambiguity. Separately, the dataset's own analysis files include, for every single word, a short raw EEG snippet that was recorded during the very first moment someone's eyes fixated on that word.

That word-sized snippet is a small, exact copy of a piece of the full sentence recording. So we can find precisely where it belongs by sliding it along the full recording and checking for a byte-for-byte match — like fitting a jigsaw-puzzle piece into the one spot where its edges line up perfectly. Once we find that spot, we know the exact instant (down to the individual EEG sample, i.e. 1/500th of a second) that word's reading began, expressed on the same clock the raw EEG file itself uses.



This method needs no clock synchronization and no guesswork — it's an exact match, so either it succeeds (and we get a precise, trustworthy timestamp) or it fails cleanly (and we skip that word rather than guess). We validated this approach on real data before trusting it for grading: across many sentences, it consistently reconstructed timings that matched independent structural checks.

We used two flavors of this same idea over the course of the project:

- **Across all 12 subjects** (the main evaluation): we reconstructed true word onset times for every subject, task, and recording block, using the method above.
- **One extra, fully self-contained example block** (subject "YHS"): a version of the same method was run ahead of time to produce a single file containing 625 words with verified ground-truth timing, bundled together with its own raw EEG. We used this as a fast, easy-to-inspect example for step-by-step debugging (Section 5), separate from the main 12-subject evaluation.

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Method 1: detecting words from the "Visual Word Form Area"

The first detector looks at one specific, small patch of scalp electrodes positioned over a brain region researchers associate with recognizing written word-shapes, sometimes called the Visual Word Form Area, or **VWFA** for short. The working idea (a hypothesis we're testing, not an established fact) is that this region's electrical activity rises and falls in a recognizable pattern whenever a word is being visually processed.

How the detector works, step by step

1. **Average a handful of nearby electrodes** into one combined signal for that brain region, to reduce noise.
2. **Smooth the signal into an "envelope"** — a curve that tracks how strong the activity is at each moment, ignoring the signal's rapid up-and-down wiggles and just showing its overall intensity.
3. **Sort every moment into one of three activity levels** — low, medium-high, or very-high — using cut-off points calculated fresh from that recording's own data (never a fixed number chosen in advance), so the thresholds adapt to each person and each recording session.
4. **Flag "medium-high" stretches that last long enough** (at least 40 milliseconds) as candidate word-reading moments, and find the strongest instant within each stretch.
5. **Thin out candidates that are too close together** — if two candidates are within a quarter of a second of each other, only the stronger one survives, since a person can't realistically be starting two new words that quickly.
6. **Draw a small time window** (0.2 seconds before to 0.2 seconds after) around each surviving candidate — that window is the detector's final answer: "a word was being read around here."

> *It's a bit like watching a seismograph and flagging every tremor that's strong enough and lasts long enough to be a real earthquake, rather than background noise — then making sure you don't report the same earthquake twice.*

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An early idea that didn't pay off: requiring a second brain region to agree

Our first working version didn't stop at Step 6 above. It added one more rule: a candidate only counted as a real detection if a *second* brain region — associated with understanding language, known as **Wernicke's area** — also showed elevated activity, in a narrow window 100–150 milliseconds after the first region's peak. The idea was that requiring two independent brain regions to agree should filter out false alarms.

What we found when we checked it word-by-word

To find out whether this "confirmation" rule was actually helping, we took one full recording block with 625 words of trustworthy ground truth and traced every single word through every stage of the detector, recording exactly where each one succeeded or failed:

WHAT HAPPENED TO THE WORD	COUNT	% OF 625
Never showed a strong-enough signal at all	26	4.2%
Signal was strong, but too brief	19	3.0%
A real candidate existed, but a nearby stronger one won	120	19.2%
Thrown out by the "second region must agree" rule	**111**	**17.8%**
Detected, but the timing window missed the word slightly	5	0.8%
Correctly detected	344	55.0%

The finding: the confirmation rule was throwing away more correctly-found words (17.8%) than the two earlier "the signal itself wasn't good enough" stages put together (7.2%). It was doing more harm than good.

Why? Two reasons. First, the 100–150ms window is quite narrow (about a twentieth of a second) and has no flexibility — if the real response from that second region happened slightly earlier or later, the rule simply missed it. Second, Wernicke's area was represented by only three scalp electrodes (the smallest, noisiest region we used), and because its own "elevated" threshold was calculated to cover about half of all its activity by definition, whether a given narrow time-window happened to land on an "elevated" stretch was often close to a coin flip — not a reliable, meaningful confirmation.

We tested removing the rule entirely and re-scored every one of the 12 subjects:



Result: removing the confirmation step improved the overall balance of correct-vs-missed detections (F1 score, a standard way of combining "how many detections were real" and "how many real words got detected" into one number) for **every one of the 12 subjects individually**, by between 3 and 11 points. Recall (the share of real words actually found) jumped from 56.5% to 72.8% overall, at a small cost to precision (the share of detections that were real words), which dropped from 60.1% to 57.7%.

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Method 2: a genuinely different way of using Wernicke's area

Removing the confirmation rule didn't mean giving up on Wernicke's area — it meant the first attempt at using it (requiring both regions to agree, on every single detection) was the wrong design. The next idea was to build a completely separate, independent detector for Wernicke's area, using a different kind of signal analysis than the VWFA detector — something called **ERP analysis**.

What is an ERP?

ERP stands for "event-related potential." When the brain processes something meaningful — like recognizing and understanding a word — it tends to produce a characteristic up-and-down wiggle in the electrical signal a fixed amount of time later, almost like a fingerprint. Researchers have named several of these wiggles based on when they occur and whether they go up or down: for instance, a dip around 300–500 milliseconds after a meaningful word (called "N400") is one of the best-known signatures of the brain doing the work of understanding a word's meaning.

> *If the VWFA method (Section 3) is like watching for a loud noise, ERP analysis is more like listening for a specific, recognizable little tune that always plays a fixed number of beats after something meaningful happens.*

How the new detector works

1. **Look at Wernicke's area signal directly** (unlike the VWFA method, we keep the signal's up/down direction intact here, since these ERP "wiggles" are defined partly by which way they point).
2. **Find prominent dips** in the signal — candidate spots where an N400-like response might be happening.
3. **Work backwards to estimate when the word actually started.** Since this kind of brain response shows up *after* the triggering word, not during it, we shift our guess for "when did the word begin" back by about 400 milliseconds (the middle of the typical 300–500ms window) from each dip we find.
4. **Double-check using multiple signatures, not just one.** Around each estimated word onset, we measure four different well-known signatures (not just the 300–500ms one), each linked to a different stage of recognizing a word, and combine their strengths into one overall "how confident are we" score.
5. **Keep only the strongest candidates** — the top 10%, by score, again using a cutoff calculated fresh from that recording rather than a fixed number.
6. **Thin out and window** the survivors exactly the same way as the VWFA method (Section 3, steps 5–6), so both detectors' final outputs are directly comparable.

This detector was built as an entirely separate program from the VWFA detector, reusing the VWFA code only by calling it (never copying or modifying it), so the two methods stay cleanly independent and the original VWFA-only results remain untouched and reproducible.

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Combining the two detectors: adding, not requiring agreement

This time, the two detectors were combined the opposite way from Section 4's failed experiment. Instead of requiring both to agree before accepting *any* detection, we let each one work independently and then simply **added together** what each one found:

- Every word the VWFA detector found stays in the final list, no questions asked.
- A word the Wernicke-ERP detector found is **added on top**, but only if it lands somewhere the VWFA detector didn't already flag — so the second method can only ever *rescue* words the first one missed. It can never cancel out or duplicate a VWFA detection.

> **In short:** Section 4's rule was an "AND" (both must agree, or the detection is thrown out). Section 6's rule is an "OR" (either one finding it is enough, and they can never take a detection away from each other).

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Results

Overall, across all 12 subjects (both reading tasks combined)

VERSION	TRUE WORDS	DETECTIONS MADE	CORRECTLY MATCHED	PRECISION	RECALL	F1
Original (VWFA + confirmation rule)	92,320	86,683	52,123	60.1%	56.5%	58.2%
VWFA only (confirmation rule removed)	92,320	116,549	67,248	57.7%	72.8%	64.4%
VWFA + Wernicke-ERP (combined)	**92,320**	**116,832**	**67,311**	**57.6%**	**72.9%**	**64.4%**

Precision answers "of everything the detector flagged, how much was actually a real word?" **Recall** answers "of all the real words, how many did the detector actually find?" **F1** is a single score that balances the two, so neither one can be gamed on its own (e.g. a detector that flags the entire recording as "word being read" would have perfect recall but terrible precision).

The combination barely moved the numbers — and we know why

> **Finding:** adding the Wernicke-ERP detector only rescued 65 extra correct words out of 92,320 total — a 0.07 percentage-point recall gain. The reason is mechanical, not a flaw in the new detector: a typical 5-minute recording block gets roughly 640 VWFA detection windows, each about 0.4 seconds wide. Laid end to end, that already covers **about 90% of the entire recording's timeline**. There's very little "empty" time left over for a second, add-only detector to contribute to.

The one place it helped noticeably was an unusually long recording block (about 42 minutes, compared to the typical 5–10 minutes) belonging to one subject — a much longer recording naturally leaves more uncovered time, and that single block alone accounted for over a third of all the extra words the combination found.

Per-subject results

SUBJECT	TRUE WORDS	VWFA-ONLY RECALL	COMBINED RECALL	RESCUED BY WERNICKE-ERP
ZAB	7,819	66.0%	66.0%	2
ZDM	7,124	69.4%	69.5%	4
ZDN	6,210	70.2%	70.3%	6
ZGW	6,855	72.3%	72.4%	3
ZJM	10,067	73.5%	73.5%	5
ZJN	10,615	83.1%	83.2%	10
ZJS	4,295	68.1%	68.3%	9
ZKB	6,650	68.6%	68.6%	3
ZKH	8,931	75.7%	75.8%	4
ZKW	10,428	79.1%	79.2%	7
ZMG	7,888	69.0%	69.1%	6
ZPH	5,438	67.2%	67.3%	6

One subject (ZJS) stands out as consistently the weakest performer across every version of the detector, largely explained by that one abnormally long recording block mentioned above, which produces a flood of low-quality detections and drags precision down for that subject specifically.

Checking the work at the individual-sentence level

Beyond the overall numbers, we also built a way to zoom into individual sentences and see, word by word, which detector (if either) caught each one — using the same strict matching rule as the headline numbers above (each detection can only be credited to one true word, never spread across several nearby words). We sampled 5 random sentences from each of the 12 subjects (60 sentences in total) and inspected them this way as a sanity check that the aggregate numbers reflect what's really happening at the word level, not an artifact of how the numbers were computed.

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What this means, and what we'd try next

Two independent, EEG-only methods for detecting word-reading events both work meaningfully better than chance, and removing an overly strict "confirmation" rule between them substantially improved results without needing any new data or hardware. The two brain regions turned out to be more useful *separately* than *jointly* in this design — requiring agreement threw away good detections, while allowing either one to contribute independently kept all the VWFA method's strengths and added a small amount more.

The Wernicke-ERP detector's contribution was real but small under the current design. Two concrete ideas, not yet tried, could change that balance:

- **Loosen how selective the Wernicke-ERP detector is** — it currently only keeps its strongest 10% of candidates by score; allowing more through would surface more candidates for the combination step to work with.
- **Let a strong Wernicke-ERP candidate compete in already-covered time**, rather than being blocked automatically whenever a VWFA detection is nearby — some of VWFA's ~640 per-block detections are themselves false alarms, and a stronger nearby Wernicke-ERP candidate might sometimes be the better answer for that particular moment.

Both of these are genuine design choices with trade-offs (they'd likely increase recall further, at some further cost to precision) rather than free improvements, and neither was tested in this round of work.

APPENDIX

Plain-language glossary

EEG (electroencephalography)

Recording electrical activity from the brain using electrodes placed on the scalp. Doesn't require surgery; the signals are relatively weak and noisy compared to more invasive methods.

Envelope

A smoothed curve tracking a signal's overall intensity over time, with the fast up-and-down wiggling removed — useful for asking "how strong is the activity right now," separate from its exact shape.

ERP (event-related potential)

A characteristic, repeatable pattern in brain activity that shows up a predictable amount of time after something meaningful happens — e.g. seeing a word, hearing a sound, recognizing a face.

VWFA (Visual Word Form Area)

A brain region associated with recognizing the visual shape of written words, used here as the source for the first, envelope-based detector.

Wernicke's area

A brain region associated with understanding language meaning, used here as the source for both the (removed) confirmation rule and the new ERP-based detector.

Ground truth

A trustworthy, independently-verified answer key used to check whether a detector's guesses are actually correct. Here, built from eye-tracking data cross-referenced against the raw EEG's own timing markers.

Precision / Recall / F1

Standard ways of scoring a detector. Precision: of the things it flagged, how many were real? Recall: of all the real things, how many did it find? F1: one number balancing both, so a detector can't cheat by only optimizing one of them.

Refractory window / non-max suppression

A rule that prevents a detector from reporting the same event twice in quick succession by keeping only the strongest candidate within a short time window.

Calibrated / percentile-based threshold

A cutoff value calculated fresh from each recording's own data (e.g. "the top 10% of values"), rather than a single fixed number chosen in advance — so the detector adapts to each person and each recording session automatically.